

Jerry (Jiarui) Hu

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EDUCATION

Sept 2026 – Present | University of Illinois Urbana-Champaign (Grainger College of Engineering)

B.S. in Computer Engineering

Prev: Sept 2021 – May 2026 | White Rock Christian Academy (WRCA) — International Baccalaureate Candidate

SKILLS

Languages: C, Java, C++, Python, Swift, SQL, HTML/CSS/JavaScript, Node.JS

Embedded & Hardware: ESP32, ROS 2, SPI, I2C, UART, STM32CubeMX, PCB Design + Soldering, 3D CAD, 3D Printing

Frameworks & Tools: KiCAD, Fusion360, Scikit-learn, NumPy, React, Firebase, Tailwind, Figma, Claude Code, Codex, Cursor

EXPERIENCE

Aug 2026 – Present | UIUC Liquid Rocketry at Illinois (LRI)

Hardware Engineer, Airframe Avionics — *STM32, KiCad, C, C++, Python (Pandas, Numpy, Pyarrow)*

- Analyzed post-launch sensor data in Python (Pandas, NumPy) for upcoming launch in November 2026
- Designed KiCAD schematics for HAL-1's barometric altimeter (3x BMP390L over SPI, per-sensor chip-select and interrupt lines) and microSD logger on an STM32H7 flight computer

Sept 2023 – June 2026 | Constellation Networking (constellationhub.org)

Chief Technology Officer (Co-founder) — *Cloudflare, Python, Firebase (Firestore, Auth, Storage, Cloud Functions)*

- Built + grew platform to 1.5k monthly visitors; recruited and led 3-person dev team as co-founder
- Built a Firebase Cloud Functions matching engine connecting students with industry professionals

Feb 2024 – June 2026 | Millennium STEM BC

Director of IT — *GitHub Pages, Firebase (Storage, Cloud Functions), TailwindCSS, HTML/JavaScript*

- Identified severe vulnerabilities on organization website; unobfuscated code and verbose logging left the org's Firestore database exposed via browser devtools; recruited as Director of IT to resolve and rebuilt website
- Built database containing 70+ STEM volunteer opportunities; hosted annual stem conference of 300+ attendees

PROJECTS & RESEARCH

Aug 2024 – Feb 2025 | National High School Data Analysis & AI Challenge

Research Team Lead — *Python, NumPy, Scikit-learn, Matplotlib, Plotly*

- Initiated research topic and methodology; led 5-person team analyzing youth substance-use trends across Canada.
- Co-authored and published research paper in STEM Fellowship Journal; advanced to STEM Fellowship Big Data & AI Challenge National Finals.

Oct 2024 | WiFi Triangulation Algorithm

Python, NumPy, Scikit-learn, Plotly 3D, Matplotlib

- Wrote a Python library to extract RSSI (proximity) data from a mesh router network.
- Trained Random Forest Model on labeled samples to predict a device's location with room-level-accuracy.

July – Aug 2025 | Smart Home Garage Door Controller

ESP32 BLE Beacon + Self-designed Relay PCB, KiCAD, Fusion360

- Designed a relay PCB to interface with garage door motor controller; integrated to open source project Home Assistant
- Built a proximity-trigger using BLE RSSI distance estimation with a BLE-beacon in car to actuate the door.

Apr – May 2025 | Motion Sensor System

ESP32-C2, PIR Sensor, KiCAD, Fusion360, MQTT, Home Assistant

- Designed custom PCBs and 3D designed + printed enclosures for 10 hand-soldered PIR motion sensors; integrated via MQTT to Home Assistant for real-time occupancy reporting.

AWARDS & LEADERSHIP

- **Honour Roll, Top 5% (Junior) + Top 25% (Senior) in Canadian Computing Competition** — 2024
- **1st place, Best Overall Hack (Major Leagues Hacking)** — Hacks for Hackers Hackathon (2024)
- **1st place, UIUC 54 Hackathons** — YC S22 Kamion: Supply Chain & Automation Track (2026)
- **2nd place, Solving Global Challenges (Yale Young Global Scholars)** — co-designed low-cost water pollution monitor bot (2024)
- **Founder, Computing Club** (grew to 21 members)